Astronomy

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Fundamentals of Astronomy

1.1 The Observer's Sky: Coordinates

To map the sky, we project positions onto the **Celestial Sphere**.

Equatorial System (Universal)

Analogous to Latitude/Longitude but fixed to stars.

- **Declination (δ):** Angle N/S of Celestial Equator. Range: $\pm 90^\circ$.
- **Right Ascension (α):** Angle E along equator from the Vernal Point (γ). Measured in Hours ($0h - 24h$).
- **Ecliptic:** The apparent path of the Sun on the sphere; tilted by 23.5° relative to the Equator.

Time Measurement

Sidereal Time: Time for Earth to rotate 360° relative to stars ($23h\,56m\,4s$).
Solar Time: Time relative to the Sun ($24h$).
Difference: Earth orbits the Sun, requiring an extra $\sim 1^\circ$ rotation per day to face the Sun again.

1.2 Telescopes and Resolving Power

Astronomy is photon-starved. Telescopes serve two main purposes:

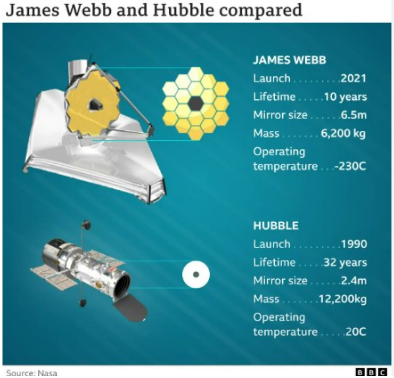
1. **Light Gathering Power:** $P \propto D^2$ (Diameter). Allows detection of fainter objects.
2. **Resolving Power:** The ability to distinguish close sources.

Rayleigh Criterion

Angular resolution θ is limited by diffraction:

$$\theta \approx 1.22 \frac{\lambda}{D}$$

- **Seeing:** Atmosphere limits ground resolution to $\sim 1''$.
- Space Telescopes (HST, JWST) reach diffraction limit.



Telescope resolution concept.[3]

1.3 Radiation Physics: The Black Body

Stars are approximated as Black Bodies. For the Sun: $T_{eff} \approx 5778$ K.

Planck's Law

Describes the spectral density $B_\lambda(T)$. Integrating this gives the laws below [6]:

1. Wien's Displacement Law

$$\lambda_{max} T = 2.898 \times 10^{-3}\,\text{m} \cdot \text{K}$$

Colors: Hot stars (O, B) are Blue; Cool stars (M) are Red.

2. Stefan-Boltzmann Law

$$L = 4\pi R^2 \sigma T_{eff}^4$$

Physics: A small increase in T yields a massive increase in Luminosity ($L \propto T^4$).

1.4 Magnitudes: The Logarithmic Scale

Formalized by [15]. Defined such that star Vega has $m \approx 0$.

Apparent Magnitude (m)

$$m_1 - m_2 = -2.5 \log_{10}(F_1/F_2)$$

The Sun: $m \approx -26.7$. Limit of human eye: $m \approx +6$. HST limit: $m \approx +30$.

Absolute Magnitude (M) and Distance Modulus

Magnitude if the object were at $d = 10$ pc.

$$m - M = 5 \log_{10}(d) - 5$$

Bolometric Magnitude: Integrates flux over all wavelengths (not just visible).

1.5 Spectroscopy: Decoding Light

- **Kirchhoff's Laws:** Hot solid $\rightarrow$ Continuous; Cool gas in front $\rightarrow$ Absorption.
- **Spectral Classification:** Cannon and Pickering (Harvard) ordered stars by Temperature:

O B A F G K M (Hot $\rightarrow$ Cold)

"Oh Be A Fine Girl/Guy, Kiss Me"

The Doppler Shift

Radial velocity changes wavelength λ:

$$z = \frac{\Delta\lambda}{\lambda_0} \approx \frac{v_r}{c}$$

Used to find Exoplanets (Radial Velocity method) and Expansion [9].

1.6 The Cosmic Distance Ladder

Overlapping methods are crucial for calibration [12].

1. **Parallax ($d < 10$ kpc):** Geometric triangulation. $d(pc) = 1/\pi('')$.
   - *Gaia Mission:* Measuring billions of stars with μas precision.
2. **Cepheid Variables ($d < 30$ Mpc):** Discovered by [11]. Luminous supergiant stars.

$$\text{Period} \propto \text{Luminosity} \quad \text{(Leavitt Law)}$$

3. **Type Ia Supernovae ($d > 1$ Gpc):** Explosion of White Dwarfs. Standardizable candles ($M \approx -19.3$). Key to Dark Energy discovery.

Stars: Structure and Evolution

2.1 The Virial Theorem and Star Formation

Stars form from the collapse of molecular clouds (ISM).

Jeans Instability

Collapse occurs when Gravity overcomes Thermal Pressure ($|E_{pot}| > 2E_{kin}$).

Timescales

Dynamical: Free fall time ($\sim 10^5$ yr).
Kelvin-Helmholtz: Time to radiate gravitational binding energy ($\sim 10^7$ yr for Sun).
Nuclear: Time to burn fuel ($\sim 10^{10}$ yr for Sun).

2.2 Radiative Transfer and Opacity

Energy must travel from the core to the surface via Radiation or Convection. The efficiency is determined by **Opacity** (κ).

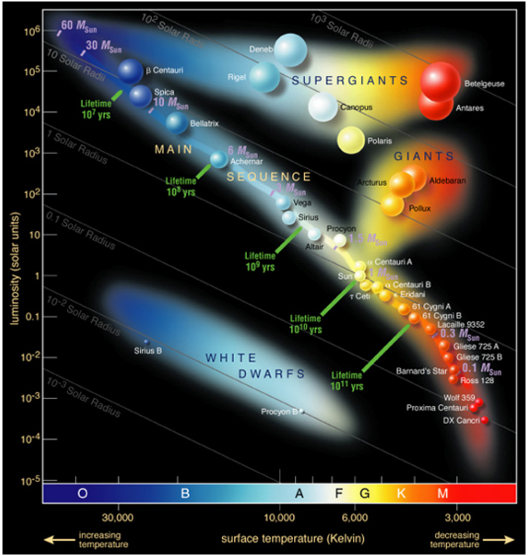
Sources of Opacity

What stops photons from escaping?

- **Bound-Bound:** Electron jumps between orbitals (Spectral Lines).
- **Bound-Free:** Photoionization.
- **Free-Free:** Electron passes near ion (Bremsstrahlung).
- **Electron Scattering:** Thomson scattering (dominant in hot, ionized interiors).

The **Optical Depth** (τ) determines how far we see: we observe photons from where $\tau \approx 1$ (Photosphere).

2.3 The Hertzsprung-Russell (HR) Diagram



Plots Luminosity vs. Temperature.

- **Main Sequence (V):** H-burning core. Mass determines position ($L \propto M^{3.5}$).
- **Giants (III):** Shell burning phases.
- **Supergiants (I):** Evolution of massive stars.
- **White Dwarfs:** Dead cores cooling down.

2.4 Energy Sources: Nucleosynthesis

Detailed in [10].

1. **Proton-Proton (pp) Chain**
   - Dominant in stars $M \lesssim 1.3M_{\odot}$.
   - $T_c \sim 15$ MK.
   - Efficiency: 0.7% of mass converted to energy ($E = mc^2$).
   - Produces Neutrinos (observed by Super-K, Borexino).

2. **CNO Cycle**
   - Dominant in massive stars ($M > 1.3M_{\odot}$).
   - C, N, O act as catalysts.
   - Reaction rate $\epsilon_{CNO} \propto \rho T^{17}$ vs $\epsilon_{pp} \propto \rho T^4$.
   - Requires higher T to overcome Coulomb barrier of C.

2.5 Stellar Populations

Walter Baade categorized stars based on their location and chemical composition (metallicity Z) [2].

	Population I	**Population II**
Age	Young (< 10 Gyr)	Old (> 10 Gyr)
Metallicity (Z)	Solar-like ($2 - 3\%$)	Metal Poor ($< 0.1\%$)
Location	Galactic Disk	Halo / Globular Clusters
Orbit	Circular, Ordered	Elliptical, Random
Color	Blue (includes O,B)	Redder

Population III: Hypothetical first generation of stars ($Z = 0$), massive and short-lived, responsible for initial chemical enrichment.

2.6 Post-Main Sequence Evolution

Destiny is determined by **Mass** [12].

Initial Mass	Final Stage	Remnant
$< 0.08M_\odot$	Brown Dwarf (No fusion)	None
$0.08 - 8M_\odot$	Red Giant $\rightarrow$ Planetary Nebula	White Dwarf
$8 - 25M_\odot$	Supernova Type II	Neutron Star
$> 25M_\odot$	Hypernova / Direct Collapse	Black Hole

Iron Catastrophe: Fe has the highest binding energy per nucleon. Fusing it requires energy (endothermic). Core collapses in $t < 1s$.

Compact Objects: The Undead Stars

3.1 White Dwarfs

- **Physics:** Supported by Quantum Electron Degeneracy Pressure. $\Delta x \Delta p \geq \hbar/2$.
- **Properties:** Mass $\sim 0.6 M_\odot$, Size $\sim R_{Earth}$.
- **Inverse Relation:** $R \propto M^{-1/3}$ (More massive WDs are smaller!).

Chandrasekhar Limit [7]

Relativistic degeneracy imposes a maximum mass:

$$M_{max} \approx 1.44 M_\odot$$

If accretion pushes $M > M_{max}$ in a binary system, C-fusion ignites explosively $\rightarrow$ **Type Ia Supernova**.

3.2 Neutron Stars and Pulsars

Remnants of Core-Collapse Supernovae [14].

- **Composition:** Superfluid Neutrons + Superconducting Protons.
- **Structure:** Mass $\sim 1.4 M_{\odot}$, Radius ~ 10 km. Density $\sim 10^{14}$ g/cm^3.

Pulsars

Discovered by Jocelyn Bell (1967). Rapidly rotating NS with strong magnetic fields ($B \sim 10^{12}$ G).

- **Lighthouse Model:** Radiation beam sweeps across Earth.
- **Precision:** Used to detect Gravitational Waves (Pulsar Timing Arrays) and test GR (Hulse-Taylor binary).

3.3 Gravitational Waves

Einstein (1916) predicted ripples in spacetime caused by accelerating masses.

- **Strain (h):** The relative change in length $\Delta L/L$. For astronomical sources, $h \sim 10^{-21}$ (size of a proton over distance to stars).
- **Sources:** Compact Binary Mergers (BH-BH, NS-NS).

LIGO Discovery [1]

On Sept 14, 2015, LIGO detected GW150914: the merger of two Black Holes ($36M_\odot + 29M_\odot$). This opened the era of **Multi-messenger Astronomy**.

3.4 Black Holes

Predicted by GR, mathematically described by [17].

Region where $v_{esc} > c$.

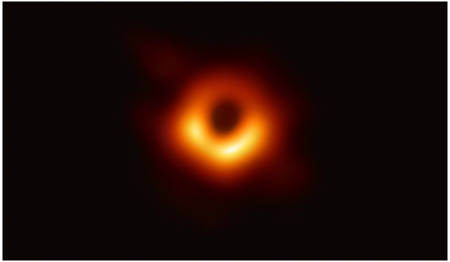
$$R_S = \frac{2GM}{c^2} \approx 3 \text{ km} \cdot \left(\frac{M}{M_\odot}\right)$$

Detection:

- X-ray Binaries (Cygnus X-1).
- Gravitational Waves (LIGO/Virgo).
- **Event Horizon Telescope:** Direct imaging of the shadow (M87*, Sgr A*).

No Hair Theorem

A BH is fully described by only 3 numbers: Mass (M), Spin (J), Charge (Q).



Simulation or EHT Image of a Black Hole.[20]

3.5 Accretion Power

The most efficient energy source (up to 40% efficiency vs 0.7% for fusion).

Eddington Luminosity

Max luminosity for a body powered by spherical accretion. Outward radiation pressure balances inward gravity:

$$L_{Edd} \approx 1.3 \times 10^{38} \left(\frac{M}{M_\odot} \right) \text{ erg/s}$$

Used to estimate the mass of Quasars ($M \sim 10^9 M_\odot$).

Galaxies and Large Scale Structure

4.1 The Milky Way

We live in a Barred Spiral Galaxy (SBc) [18].

- **Components:**
  - *Thin Disk:* Young stars (Pop I), gas, dust. Site of star formation.
  - *Thick Disk/Bulge:* Older stars, metal-poor.
  - *Halo:* Globular Clusters, old stars (Pop II), Dark Matter.
- **Center:** Sagittarius A* $(4 \times 10^6 M_\odot)$. Orbits of S2 star confirm GR predictions.

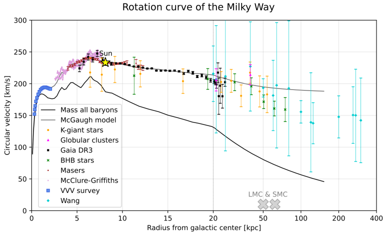
Structure of the Milky Way. [13, 4]

4.2 The Dark Matter Problem

Visible mass (baryons) is insufficient to explain dynamics [5].

Evidence

1. **Rotation Curves:** Flat at large radii [16]. Implies halo mass $M(r) \propto r$.
2. **Gravitational Lensing:** Mass bends light more than visible matter allows.
3. **Bullet Cluster:** Collision where gas (X-rays) separates from mass (Lensing), proving DM exists as collisionless fluid.



Galaxy rotation curves evidence. [16, 19]

4.3 Active Galactic Nuclei (AGN)

Galactic centers outshining the host galaxy.

- **Unified Model:** Differences depends on viewing angle relative to a dusty torus.
    - *Type I:* Face-on (Broad lines visible).
    - *Type II:* Edge-on (Broad lines obscured).
    - *Blazar:* Jet points to Earth.
- **Feedback:** AGN jets heat the surrounding gas, stopping star formation (Quenching).

4.4 Galaxy Evolution and Mergers

Galaxies are not static; they evolve through the **Hierarchical Model**.

Mergers:

- **Major Merger:** Collision of two equal-sized spirals often results in an Elliptical Galaxy (scrambling of orbits).
- **Minor Merger:** A large galaxy absorbs a smaller one (Galactic Cannibalism), often fueling the AGN or star formation bursts.

Future of Milky Way:

- On collision course with **Andromeda (M31)**.
- Merger expected in ~ 4.5 Gyr.
- Result: "Milkomeda" (likely an Elliptical Galaxy).

4.5 Gamma Ray Bursts (GRBs) and Kilonovae

Most energetic electromagnetic events since Big Bang.

Long GRBs (> 2s)

- Collapse of massive rotating star ("Collapsar").
- Associated with Type Ic Supernovae.

Short GRBs (< 2s)

- Merger of two Neutron Stars.
- **GW170817:** Historic event detected in GW and Light.
- **Kilonova:** Production of heavy r-process elements (Gold, Platinum).

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